

# MODULE 4: PRACTICE PYTHON FUNDAMENTALS

Practice using these Python building blocks to start writing readable, structured scripts.

## Create a New Python File

1. In VS Code, click File > Open Folder, and create/select a folder like python-course
2. In the Explorer tab (left sidebar), click New File and name it: **basics.py**

## Practice fundamental tasks

Type the following text into the terminal and observe the outputs.

| DEFINITIONS                                                                                                           | CODE                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| A <b>VARIABLE</b> stores information (like a name or number) so you can reuse it later.                               | <pre>name = "Sam" # a string (text) age = 34 # an integer (number) is_writer = True # a boolean (True or False)</pre>                                  |
| Use <b>PRINT()</b> to show the result of your code in the terminal.                                                   | <pre>print("Name:", name) print("Writer?", is_writer)</pre> <p><b>The output will look like this:</b><br/>Name: Sam<br/>Writer? True</p>               |
| Use <b>+</b> for basic combinations of text and variables. (Should be used rarely)                                    | <pre>print(name + " is a technical writer.")</pre>                                                                                                     |
| Use <b>F-STRING</b> for all other combinations of text and variables. (Best practice)                                 | <pre>print(f"{name} is a technical writer aged {age}.")</pre>                                                                                          |
| Include <b>MATH SYMBOLS</b> for basic equations.<br><br>+ addition<br>- subtraction<br>* multiplication<br>/ division | <pre>docs_written = 10 days = 5  print("Docs per day:", docs_written / days)</pre> <p><b>The output will look like this:</b><br/>Docs per day: 2.0</p> |
| A <b>COMMENT</b> starts with <b>#</b> and is ignored by Python. It's just for humans reading the code.                | <pre># This script prints a summary for a technical writer</pre>                                                                                       |